Lecture No 1

Man-made Hazards:

Disasters have been a part of human life since the beginning of time. Man-made hazards and catastrophes are events that reduces quality of life, health issues, and even fatalities. Some natural catastrophes have become more frequent and intense as a result of human influences.

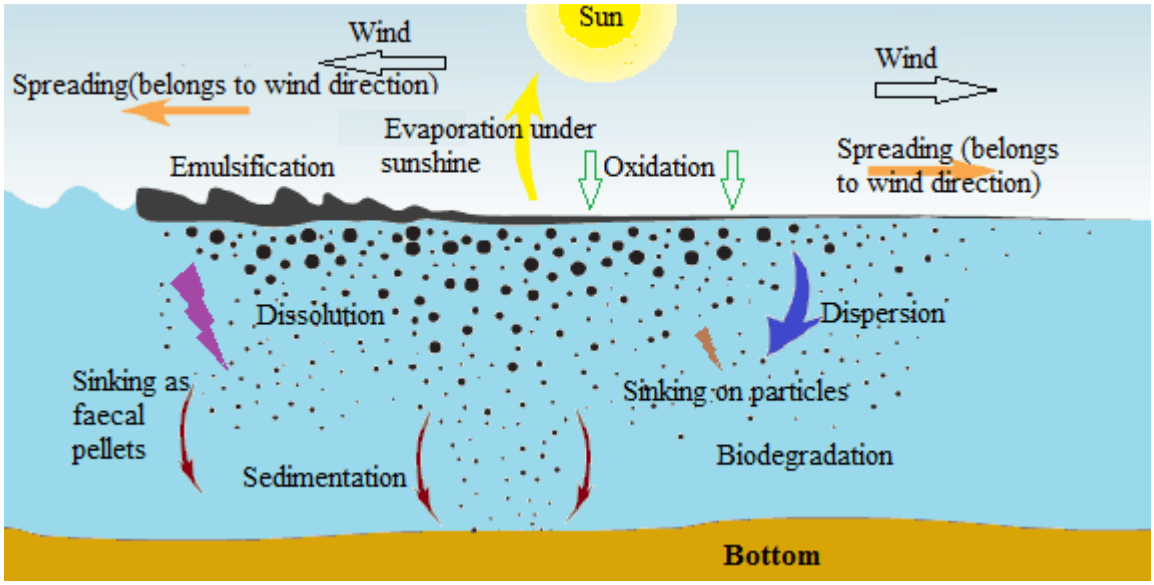
- Man-made disasters can be intentional, by accident or due to improper management (negligence) of man made systems.
- Avoiding man-made hazards requires caution from authorities and proper disaster management plans at a local level. When members of human communities intend to inflict damage on fellow members, strict surveillance mechanisms help to avert danger.
- A hazard exists only when it affects a person exposed to it. But not all disasters come with an alarm bell.

Definition of Man-made Hazards

- Man-made hazards are **disruptions of a massive scale** over a short or long period of time, caused by **human action or inaction**.
- These disasters cause economic damage, loss of life, health emergencies, etc.
- Man-made disasters can be classified into Sudden disasters, Continuing disasters, and Armed conflicts.
- Human forces are more responsible for **sudden disasters** than natural ones. Sudden hazards include the discharge of methyl isocyanate from the Union Carbide factory in Bhopal in 1984 and the 1986 emission of radioactive chemicals from the Chernobyl nuclear plant in the Soviet Union. Terrorist attacks unleashed on citizens are also sudden disasters.
- Oil spills and pollution over a long period of time are examples of continuing man-made hazards and disasters. The causal factor cannot be blocked immediately leading to a lasting impact.
- **Wars and armed conflicts** have constantly mired human history. There are domestic upheavals, territorial conflicts between countries, and intervention of extra state entities like terrorist organizations

Types of Man-made Hazards-

Oil and Chemical Spill



- People, animals, land and the marine ecosystem are all affected by an oil and chemical spill, making it one of the most destructive environmental man-made hazards that may occur.
- Such accidental spills harm and pollute vital resources in the food supply system, among other things.
- Explosives, flammable and combustible materials, poisons, and radioactive materials are examples of hazardous materials. Accidents during transportation and chemical leakages at industrial units or refineries are the most common sources of these releases.
- Most industrial accidents may be prevented by implementing and following adequate safety standards and processes. Accidents generate harmful gases and poisonous chemicals that damage living forms in neighbouring regions.
- According to estimates, Iraqi forces fled from Kuwait in **1991** and broke valves of oil wells and pipelines, causing the **Gulf War oil leak** releasing 8 million barrels of oil into the Persian gulf. A five-inch-thick oil slick covered 101 miles by 42 miles at its peak.
- The Bhopal Gas Tragedy of 1984 was the world's worst chemical (industrial) catastrophe in 1984. Methyl isocyanate, a poisonous chemical, was accidentally released, resulting in the death of nearly 2500 individuals.

Terrorism

- To intimidate, coerce, or demand ransom, terrorists employ force or violence against people or property in violation of criminal laws.
- In order to convince individuals that their government is helpless to combat terrorism, terrorists often use threats to generate fear among the population.
- From terrorist threats to assassinations to kidnappings and hijackings, bomb scares and bombings, and cyber warfare to the use of chemical, biological and nuclear weapons, terrorism may take many forms.
- The **9/11 terrorist attacks** that took place on the Twin Towers (New York) in USA was the turning point in counter terrorism policy. The United States took a lead in suppressing the growth of terrorist networks in Iraq and Afghanistan.

- In November 2008, Mumbai, Maharashtra, India, was the scene of numerous terrorist operations. Lashkar-e-Taiba, a Pakistani terrorist group, carried out the assaults in Mumbai. A number of landmarks in Mumbai's south were attacked by terrorists, including the Chhatrapati Shivaji train station and Leopold Café, two hospitals, and a theatre.

Wars

- Wars are fully fledged armed attacks between states that wreak immense havoc to life. It leads to huge economic losses, deaths and psychological distress.
- The two World Wars of the 20th century saw millions lose their life and loved ones.
- Wars are usually the culminations of long standing rifts between countries that spiral out due to trigger events.
- **Nuclear bombing of Hiroshima and Nagasaki** on **6th and 9th August of 1945** wreaked unforeseen damage on the cities.
- The form and platform for wars have shifted from being geographical-physical conflicts to cyberspace. Governments and extra state actors have the ability to disrupt normal functioning of lives by attacking important facilities like power plants, refineries etc.
- The Kargil War of 1999 was fought between India and Pakistan along the Line of Control. Hundreds of soldiers from both sides succumbed to death.

Nuclear Accidents

- The International Atomic Energy Agency (IAEA) defines a nuclear accident as an incident that has resulted in severe implications for the people, the environment, or the nuclear plant. Radioactive fallout and reactor core melt are potentially high damage events.
- The **Chernobyl disaster in 1986 and the Fukushima Daiichi accident in 2011** highlighted the consequences of any mishap in nuclear power plants.

Human Accelerated Hazards

- Environmental issues like pollution, climate change, ozone layer depletion are intensified due to human intervention.
- Landslides and meteorological phenomena like tornadoes, storms are classified under natural disasters. But several factors like construction of roads, destruction of green cover etc can accelerate these natural disasters. More than 5000 lost their lives in the **Uttarakhand floods of 2013**.

What is Soil Erosion?

In this process, the soil particles are loosened or washed away in the valleys, oceans, rivers, streams or far away lands. This has been worsening due to human activities such as agriculture and deforestation.

Soil erosion is a continuous process that occurs either slowly or at an alarming rate. It results in a continuous loss of topsoil, ecological degradation, soil collapse, etc.

Let us have a detailed look at the causes, effects and prevention of soil erosion.

Cause of Soil Erosion

Following are the important causes of soil erosion:

Rainfall and Flooding

Higher intensity of rainstorms is the main cause of soil erosion. Four types of soil erosion are caused by rainfall:

- Rill erosion
- Gully erosion
- Sheet erosion
- Splash erosion

The raindrops disperse the soil, which is then washed away into the nearby streams and rivers. Regions with very heavy and frequent rainfall face a large amount of soil loss. The flowing water during floods also erodes a lot of soil by creating potholes, rock-cut basins, etc.

Agriculture

The farming practices are the major cause of soil erosion. The agricultural activities disturb the ground. The trees are cleared and the land is ploughed to sow new seeds. Since most of the crops are grown during the spring season, the land lies fallow during winters. Most of the soil is eroded during winters.

Also, the tyres of tractors make grooves on the land, making a natural pathway for water. Fine soil particles are eroded by wind.

Grazing

The grazing animals feed on the grasses and remove the vegetation from the land. Their hooves churn up the soil. They also pull out plants by their roots. This loosens the soil and makes it more prone to erosion.

Logging and Mining

A large number of trees are cut down to carry out the logging process. Trees hold the soil firmly. The canopy of the trees protects the soil from heavy rainfall. The leaf litter that protects the soil from erosion, is also lost during logging.

Mining activities also disturb the land and leave the soil more prone to erosion.

Construction

The construction of roads and buildings exposes the soil to erosion. The forests and grasslands are cleared for construction purposes, which exposes the soil making it vulnerable to erosion.

Rivers and Streams

The flowing rivers and streams carry away the soil particles leading to a V-shaped erosion activity.

Heavy Winds

During dry weather or in the semi-arid regions, the minute soil particles are carried away by the wind to faraway lands. This degrades the soil and results in desertification.

Also Read: Types of Pollution

Effects of Soil Erosion

The major effects of soil erosion include:

Loss of Arable Land

Soil erosion removes the top fertile layer of the soil. This layer is rich in the essential nutrients required by the plants and the soil. The degraded soil does not support crop production and leads to low crop productivity.

Clogging of Waterways

The agricultural soil contains pesticides, insecticides, fertilizers, and several other chemicals. This pollutes the water bodies where the soil flows.

The sediments accumulate in the water and raise the water levels resulting in flooding.

Air Pollution

The dust particles merge in the air, resulting in air pollution. Some of the toxic substances such as pesticides and petroleum can be extremely hazardous when inhaled. The dust plumes from the arid and semi-arid regions cause widespread pollution when the winds move.

Desertification

Soil erosion is a major factor for desertification. It transforms the habitable regions into deserts. Deforestation and destructive use of land worsens the situation. This also leads to loss of biodiversity, degradation of the soil, and alteration in the ecosystem.

Destruction of Infrastructure

The accumulation of soil sediments in dams and along the banks can reduce their efficiency. Thus, it affects infrastructural projects such as dams, embankments, and drainage.

Also Read: Difference between erosion and weathering

Soil Erosion Prevention

Soil erosion is a serious environmental issue. Steps should be taken to curb this problem. Following are some of the methods of soil erosion prevention:

1. Plant trees on barren lands to limit erosion of soil.
2. Add mulch and rocks to prevent the plants and grass underneath to prevent soil erosion.
3. Mulch matting can be used to reduce erosion on slopes.
4. Put a series of fibre logs to prevent any water or soil from washing away.
5. A wall at the base of the slope can help in preventing the soil from eroding.
6. Every household should have a proper drainage system so that water flows down into proper water collecting systems.

Key Points of Soil Erosion

- It is the natural process of wearing away topsoil, but human activities have accelerated the process.
- It is usually caused due to the removal of vegetation, or any activity that renders the ground dry.
- Farming, grazing, mining, construction and recreational activities are some of the causes of soil erosion.

- The effects of soil erosion are not just land degradation. It has led to a drastic increase in pollution and sedimentation in rivers that clogs the water bodies resulting in a decline in the population of aquatic organisms.
- Degraded lands lose the water holding capacity resulting in floods.

The health of the soil is of utmost importance to the farmers and the population that depends upon agriculture for food and employment. There are several challenges to resist soil erosion, but there are solutions to prevent it as well.

Mechanics of Soil Erosion

The type of soil depreciation in which the uppermost layer of the soil is removed by various natural processes is widely known as Soil Erosion. Let us discuss the various mechanical tools and mechanics of the mechanism of soil erosion pdf.

Soil Erosion is one of the major problems faced by many countries. A major part of the total land available to us is lost due to cultivation, while a large gets depreciated every year due to Water Erosion and Wind Erosion. Soil Erosion is a natural process of soil depreciation. It depreciates the quality, structure, salinization, and organic matters of the soil. Topsoil has the highest quantity of organic components, composed of mineral particles, water, air, etc. Due to Soil Erosion, the topsoil gets eroded and removed among all other layers of the soil.
Soil erosion composes of three parts- detachment, movement, and deposition. Detachment is when the topsoil is detached from the other layers of the soil. Movement is the process of the topsoil moving into another region. And, Deposition is the outcome of Movement- where the topsoil deposits itself after movement.
One must keep in mind that **Soil Erosion** is not always Natural, sometimes it is the result of human degradation of forests, mismanagement, etc.

Water Erosion
Soil Erosion of Water is mainly caused by Rainfall, which is natural. However, it has two sources, namely, raindrops and surface flow of water. Raindrops fall from the sky, hence, this source is a vertical source of water erosion. While, the surface flow flows horizontally- causing the topsoil to move and deposit in another region. However, the raindrops detach, move, and deposit the topsoil into another region. **Soil Erosion** in this form has various forms. These forms are:

- *Rill Erosion:* Rill Erosions channel the detachment and movement of the topsoil, and smoothen out the raindrop glow onto the ground.

- *Splash Erosion:* Splash Erosion is the process in which the soil particles get removed because of Rain Drops.
- *Gully Erosion:* Gully Erosions are the form of soil erosion in which Gullies emerge in the grounds due to the heavy rainfall, speedy flow of water on the ground. Gullies are so deep in the surface level that they can divide the piece of land into different portions.
- *Ravines Erosion:* Ravines Erosion is the upgraded version of Gully Erosion. It is deeper than the Gullies formed due to heavy rainfall, the speedy flow of water on the ground.
- *Slip Erosion:* Slip Erosions are common in places with high altitude mountains and heavy rainfall, causing sloppy lands. The ultimate effect of these forms of erosion is an Earthquake or very heavy rainfall.
- *Stream Bank Erosion:* Stream Bank Erosion takes place when the water stream changes its direction and cuts the soil, making caves in the swollen river banks.

Wind Erosion

The arid and semi-arid regions are the most commonly affected regions of Erosion by Wind. These regions commonly do not have any kind of vegetation and the velocity of the wind is also high. Wind Erosion happens in such a way that the displaced soil forms a dune. This happens due to the high velocity of the wind in the region. The formation of these dunes makes the ground unfavourable for farming and cultivation.

The erosion of soil due to wind causes three different types of movement of Soil. These are-

- *Suspension:* The movement or transportation of tiny dust particles in the air is known as Suspension. These tiny particles are often smaller than 0.1 millimetres in diameter.
- *Saltation:* When the soil moves in a series of small bounces by the Wind then it is known as Saltation. This is the most common form of Soil Erosion by Wind. These collusion particles are 0.1 to 0.5 millimetres in diameter.
- *Surface Creep:* Due to the pressure that creates during Saltation, the huge soil particles slide and rolls along the surface, known as Surface Creep. These heavy particles are 0.5 to 3.0 millimetres in diameter.

Mechanical Control

There are various ways of **mechanical control of Soil Erosion**. These ways include Graded Bunds, Bench Terracing, Trenches, among others.

- *Graded Bunds:* These bunds are constructed in areas having heavy to average annual rainfall with 2 to 6 per cent sloping in its structure. These bunds have embankments built in the lower end of the channel through which the soil evacuates.

- *Bench Terracing:* This method consists of the building of step fields along contours with half-filling and cutting. It is the most popular method used. These are built-in lands with sloping between 16 to 33 per cent.

Trenches: These are adopted regions having heavy rainfall and slopping of more than 33 per cent. In the trench side of the bunds, various kinds of plantation in done.

Lecture No 2

Soil erosion is caused by a number of factors, including natural and human-made causes:

- Natural factors
    - Climate: Extreme temperatures, rainfall, and droughts can all contribute to soil erosion.
    - Wind: Winds of 13 miles per hour or more at one foot above the ground can erode soil.
    - Wildfires: Can cause soil erosion.
- Human-made factors
    - Agriculture: Tilling, overgrazing, and poor farm management can all lead to soil erosion.
    - Deforestation: Clearcutting and other logging practices can leave soil exposed to wind and rain.
    - Construction: Building roads and other structures can contribute to soil erosion.
    - Mining: Mining and quarrying can cause soil erosion.

Other factors that can affect soil erosion include:

- Soil erodibility
- Slope gradient and length
- Cropping and vegetation
- Tillage practices
- Soil cloddiness
- Surface roughness
- Field size
- Soil moisture

Soil erosion can lead to a number of problems, including topsoil loss, water pollution, and biodiversity loss.

Soil erosion can be caused by a number of factors, including:

- Water

  Water erosion is a primary cause of soil erosion. When it rains, raindrops break the bonds that hold soil together. The impact of the raindrops loosens the soil, and if it rains heavily, the water pools on the ground and flows, carrying the soil with it. This process is called surface water runoff.

- Wind

  Wind can blow away small particles of soil that are splashed into the air by raindrops. This is known as splash erosion.

- Human activities

  Deforestation, overgrazing, forest fires, and poor farming techniques can all contribute to soil erosion.

- Land use, climate, and slope

  These factors can contribute to gully erosion, which is an intense form of soil erosion that creates open channels in the ground.

- Sheet erosion
  This occurs when water flows as a sheet over a large area down a slope, washing away the top soil.

Soil erosion can lead to desertification, where habitable land is slowly converted into desert. It can also pollute nearby water bodies and wetlands, and reduce the productivity of cropland

Here are some conservation measures to prevent soil erosion:

- Strip cropping

  Most effective when placed along the land's contour to control erosion on slopes. To control wind erosion, place the strips at right angles to the prevailing winds.

- Cover crops

  Helps retain soil, reduce erosion, and slow rainwater runoff. Cover crops also reduce soil compaction, suppress weed growth, and reduce nutrient leaching.

- Crop rotation

  Improves soil structure and reduces the amount of soil exposed to wind and water. Crop rotation also supports reduced or no-till farming, which provides even better protection against erosion.

- Conservation tillage

  A strategy to conserve soil by reducing erosion. The goal is to leave at least 30% crop residue on the soil surface.

- Terrace farming

  An efficient solution for hilly farmlands that prevents soil erosion and contributes to soil conservation.

- Windbreaks

  A field windbreak controls soil erosion and prevents crop damage and loss caused by wind.

- Mulching
  Protects soil from direct rays of the sun and moving air, which greatly reduces evaporation of water from the soil. Rain falling on the mulch prevents erosion and a compacted, crusted soil surface

Lecture No 3

Chemical disasters may be traumatic in their impacts on human beings and have resulted in huge casualties and also large-scale damage to nature and property. The elements which are at highest risks due to chemical disaster primarily include the industrial plant, its employees & workers, hazardous chemicals vehicles, the residents of nearby settlements, adjacent buildings, occupants and surrounding community.

Brief Overview

In the year 1984, India had witnessed the world's worst chemical (industrial) disaster – the "Bhopal Gas Tragedy" (which happened on December 2nd, 1984). The Bhopal Gas tragedy was the most traumatic chemical accident in history. The tragedy led to the death of over 2500 people due to the accidental release of toxic gas Methyl IsoCyanate (MIC). The current generation continues to bear the ill-effects of the tragedy.

Such accidents are significant in terms of loss of lives, suffering, pain, injuries, damage to environment and property and loss of lives. India continued to witness a series of chemical accidents even after Bhopal had demonstrated the vulnerability of the country.

There are about 1861 Major Accident Hazard (MAH) units, spread across 298 districts and 25 states & 3 Union Territories, in all zones of the country. Besides, there are thousands of registered and hazardous factories and unorganized sectors dealing with numerous ranges of hazardous material posing serious and complex levels of disaster risks.

Factors Causing Chemical Disasters

1. Ageing of process plants, defects in design and inadequate steps to pace with modern technologies in the Indian chemical industry has increased vulnerability to chemical disasters.
2. Organic solvents are the most common source of fires and explosions in the chemical industry.
3. A majority of the industrial accidents occur due to human error as a result of non-compliance of Standard Operating Procedures (SOPs). Piper alpha accident is a classic example of how human error can lead to chemical disasters, wherein a worker accidentally activates a pump under maintenance without a safety valve in place that leads to a gas leak and subsequent explosion.
4. There is an increased threat due to terrorist activities and sabotages. Natural disasters such as floods and earthquakes have also caused a major disaster in the chemical industry.
5. Another common cause that results in chemical disasters is improper maintenance of equipment. Regular maintenance at scheduled intervals following the manufacturer's recommendations is important for ensuring that the equipment runs smoothly and

safely. The Flixborough incident is a prime example of improper maintenance which led to the death of 28 people and injured many.

6. Non-availability of an emergency response team to mitigate accidents during the transportation of hazardous chemicals have also resulted in major disasters in several locations in India.
7. Hazardous waste processing and its proper disposal need special attention as these activities can also contribute to fire, explosions and toxic releases to the environment.
8. Terrorist attack/ unrest leading to sabotage.

Chemical hazards are systems that may lead to chemical accidents, which are the unintentional or intentional release of hazardous substances that can harm people, the environment, or both:

- Causes

  Chemical accidents can be caused by natural events, human error, or deliberate acts. Examples include industrial accidents, terrorist attacks, or volcanic eruptions.

- Types

  Chemical accidents can range from small releases to major emergencies, and can be sudden or have a slow onset. They can include fires, explosions, leaks, or contamination of the food or water supply.

- Impact

  Chemical accidents can cause illness, injury, disability, or death. They can also damage property and nature.

- Prevention

  Efforts to prevent chemical accidents include improved safety systems, changes in chemical use and manufacture, and monitoring of toxic waste disposal.

- Response

  If you are in a vehicle during a chemical accident, you should stop and seek shelter in a permanent building. If you must remain in your car, you should close the windows and vents, and shut off the air conditioner and heater.

The most significant chemical accident in history was the 1984 Bhopal disaster in India, where over 3,000 people died after a toxic gas was released at a pesticides factory

Toxic chemicals released by industries this year, tons
7,593,146

Sources and methods:

Facts

- Each and **every second 310 Kg of toxic chemicals are released** into our air, land, and water by industrial facilities around the world.

- This amounts to approximately **10 million tons** (**over 21 billion pounds**) of toxic chemicals released into our environment by industries each year.

- Of these, over **2 million tons** (**over 4.5 billion pounds**) **per year** are **recognized carcinogens.** This amounts to about **65 Kg** each second.

Definitions

- **toxic chemicals:** substances that can cause severe illness, poisoning, birth defects, disease, or death when ingested, inhaled, or absorbed by living organisms.

- **recognized carcinogens:** as defined by the Occupational Safety & Health Administration (view OSHA Carcinogens list)

- **release:** any spilling, leaking, pumping, pouring, emitting, emptying, discharging, injecting, escaping, leaching, dumping or disposing into the environment *(as defined by EPCRA, U.S. Emergency Planning and Community Right-to-Know Act, section, 329-8)*

- **environment**: includes water, air and land and the interrelationship which exists among and between water, air, and land and all living things.

A "release" refers to different ways that toxic chemicals from industrial facilities enter the air, water and land. Releases include spilling, leaking, pumping, pouring, emitting, emptying, discharging, injecting, escaping, leaching, dumping, or disposing into the environment.

A toxic chemical release is when toxic chemicals enter the environment through various means, such as leaking, spilling, or discharging. These releases can occur in the air, water, or land.

Type of release	**Description**
Air	Releases from point sources, like smokestacks, or non-point sources, like leaks
Water	Discharges to bodies of water, like rivers and streams, or runoff
Land	Disposal in landfills, surface impoundments, or underground injection wells

Off-site	Transfers to a facility separate from the reporting facility for disposal or other release

Toxic chemical releases can be caused by natural events, accidents, or intentional acts. They can range from small releases to major emergencies, and can have a slow onset or be sudden and acute.

Some of the chemicals released in the largest quantities by industry include:

- Chemical manufacturing: Ammonia and ethylene
- Paper manufacturing: Methanol
- Electric utilities: Sulfuric acid

If you are exposed to toxic chemicals, you should:

- Rinse your eyes with clean, cool water for several minutes
- Seek medical help
- Avoid spreading chemicals from your clothes onto your skin
- Put contaminated clothes into a closed container
- Put a warning on the bag and put it in a secure place outside until it can be disposed of

A nuclear explosion is a result of a nuclear device detonating and releasing energy rapidly. The effects of a nuclear explosion include:

- Blast: A large fireball is created that vaporizes everything inside it, including soil and water.
- Heat: The explosion produces a wave of intense heat.
- Radiation: The explosion produces a wave of radiation.
- Fallout: Radioactive material mixes with the vaporized material and falls back to Earth as particles, which can contaminate people, food, and water. Fallout is most dangerous in the first few hours after the explosion.
- Mushroom cloud: The vaporized material rises and forms a mushroom cloud.

Some notable nuclear explosions and accidents include:

- Hiroshima and Nagasaki

  The United States dropped atomic bombs on these Japanese cities at the end of World War II, killing around 109,000 people.

- Chernobyl

This accident in Ukraine in 1986 killed around 30 people and damaged billions of dollars of property. The World Health Organization estimates that up to 4,000 additional cancer deaths may be related to the accident.

- Castle Bravo
  This hydrogen bomb test in the Marshall Islands in 1954 was the worst radiological disaster in the United States' testing history.

If you are in the area of a nuclear explosion, you should:

- Find a shelter in the middle of a large building or underground, such as a basement or subway.
- Stay inside for at least 24 hours, unless told otherwise by authorities.
- Turn off fans, air conditioners, and forced-air heating units.
- Close windows, doors, and fireplace dampers.
- Follow instructions from emergency response officials

- **Nuclear Disaster**
- A nuclear and radiation accident is defined by the International Atomic Energy Agency as "an event that has led to significant consequences to people, the environment or the facility. Examples include lethal effects to individuals, large radioactivity release to the environment, or reactor core melt."The prime example of a "major nuclear accident" is one in which a reactor core is damaged and significant amounts of radiation are released, such as in the Chernobyl Disaster in 1986. The impact of nuclear accidents has been a topic of debate practically since the first nuclear reactors were constructed. It has also been a key factor in public concern about nuclear facilities.Some technical measures to reduce the risk of accidents or to minimize the amount of radioactivity released to the environment have been adopted. Despite the use of such measures, "there have been many accidents with varying impacts as well near misses and incidents".
- **Impact**
- Benjamin K. Sovacool has reported that worldwide there have been 99 accidents at nuclear power plants.[5] Fifty-seven accidents have occurred since the Chernobyl disaster, and 57% (56 out of 99) of all nuclear-related accidents have occurred in the USA.Serious nuclear power plant accidents include the Fukushima Daiichi nuclear disaster (2011), Chernobyl disaster (1986), Three Mile Island accident (1979), and the SL-1 accident (1961).Stuart Arm states, "apart from Chernobyl, no nuclear workers or members of the public have ever died as a result of exposure to radiation due to a commercial nuclear reactor incident."Nuclear-powered submarine mishaps include the K-19 reactor accident (1961),the K-27 reactor accident (1968),and the K-431 reactor

accident (1985).Serious radiation accidents include the Kyshtym disaster,Windscale fire, radiotherapy accident in Costa Rica,radiotherapy accident in Zaragoza,radiation accident in Morocco,Goiania accident,radiation accident in Mexico City, radiotherapy unit accident in Thailand,and the Mayapuri radiological accident in India.

- **Nuclear power plant accidents**
- One of the worst nuclear accidents to date was the Chernobyl disaster which occurred in 1986 in Ukraine. That accident killed 30 people directly, as well as damaging approximately $7 billion of property. A study published in 2005 estimates that there will eventually be up to 4,000 additional cancer deaths related to the accident among those exposed to significant radiation levels.Radioactive fallout from the accident was concentrated in areas of Belarus, Ukraine and Russia. Approximately 350,000 people were forcibly resettled away from these areas soon after the accident. Benjamin K. Sovacool has reported that worldwide there have been 99 accidents at nuclear power plants from 1952 to 2009 (defined as incidents that either resulted in the loss of human life or more than US$50,000 of property damage, the amount the US federal government uses to define major energy accidents that must be reported), totaling US$20.5 billion in property damages.Fifty-seven accidents have occurred since the Chernobyl disaster, and almost two-thirds (56 out of 99) of all nuclear-related accidents have occurred in the USA. There have been comparatively few fatalities associated with nuclear power plant accidents. Nuclear reactors become preferred targets during military conflict and, over the past three decades, have been repeatedly attacked during military air strikes, occupations, invasions and campaigns.
- **Nuclear Safety**
- Nuclear safety covers the actions taken to prevent nuclear and radiation accidents or to limit their consequences. This covers nuclear power plants as well as all other nuclear facilities, the transportation of nuclear materials, and the use and storage of nuclear materials for medical, power, industry, and military uses. The nuclear power industry has improved the safety and performance of reactors, and has proposed new safer (but generally untested) reactor designs but there is no guarantee that the reactors will be designed, built and operated correctly.Mistakes do occur and the designers of reactors at Fukushima in Japan did not anticipate that a tsunami generated by an earthquake would disable the backup systems that were supposed to stabilize the reactor after the earthquake.According to UBS AG, the Fukushima nuclear accidents have cast doubt on whether even an advanced economy like Japan can master nuclear safety. Catastrophic scenarios involving terrorist attacks are also conceivable. An interdisciplinary team from MIT have estimated that given the expected growth of nuclear power from 2005 – 2055, at least four serious nuclear accidents would be expected in that period.To date, there have been five serious accidents (core damage) in the world since 1970 (one at Three Mile Island in 1979; one at Chernobyl in 1986;

and three at Fukushima-Daiichi in 2011), corresponding to the beginning of the operation of generation II reactors. This leads to on average one serious accident happening every eight years worldwide. Nuclear weapon safety, as well as the safety of military research involving nuclear materials, is generally handled by agencies different from those that oversee civilian safety, for various reasons, including secrecy. There are ongoing concerns about terrorist groups acquiring nuclear bomb-making material.

Lecture No 4

Sedimentation can cause a number of global problems, including:

- Water supply

  Sedimentation can reduce the amount of water that can be stored in reservoirs, which can threaten the sustainability of water supplies.

- Hydropower

  Sediment can damage hydroelectric turbines and other dam components, which can increase maintenance costs and decrease efficiency.

- Flooding

  Sediment can fill up storm drains and catch basins, which can lead to flooding when it rains.

- Water quality

  Sediment can make water cloudy, which can make it difficult for aquatic life to see food. It can also increase the cost of treating drinking water and cause odor and taste problems.

- Wildlife habitats

  Sediment can disrupt the natural food chain and destroy the habitats of small stream organisms, which can lead to declines in fish populations.

- Recreational access

  Sediment can make reservoirs shallower, which can reduce recreational access.

Sedimentation rates and associated storage losses can vary significantly between regions. For example, one study estimated that 120 California reservoirs had lost over 75% of their original capacity. Another study predicted that Japan's Sakuma reservoir will lose around 44% of its initial capacity by 2040.

To mitigate sedimentation, planners, developers, and operators can consider how to prevent, manage, and reduce sedimentation in water storage projects.

Although sedimentation of the world’s reservoirs represents a serious threat to the sustainability of hydropower, there is limited guidance on how best to address the problem. Sedimentation affects the safety of dams and reduces energy production, storage, discharge capacity and flood attenuation capabilities. It increases loads on the dam and gates, damages mechanical equipment and creates a wide range of environmental impacts. This article explores sedimentation issues as they pertain to hydropower facilities, dam safety and the environment; discusses sedimentation management techniques; and describes how they can be implemented to limit the impacts on hydropower.

Background

Reservoir sedimentation is a process of erosion, entrainment, transportation, deposition and compaction of sediment carried into reservoirs formed and contained by dams. In

unregulated, mature rivers with stable catchments, sediment processes are relatively balanced. Construction of a dam decreases flow velocities, initiating or accelerating sedimentation,1 resulting in progressively finer materials being deposited (see Figure 1).

There are three stages in a reservoir's life:[2]

- Continuous and rapidly occurring sediment accumulation;
- Partial sediment balance, where often fine sediments reach a balance but coarse sediments continue to accumulate; and
- Full sediment balance, with sediment inflow and outflow equal for all particle sizes.

Most of the world's reservoirs are in the continuous accumulation stage.[2] Many were designed by estimating sedimentation rates in order to provide a pool with sufficient volume to achieve a specified design life. However, this design life is typically far less than what is actually achievable. Therefore, managing reservoirs to achieve a full sediment balance is essential in order to maximize their lives.

Developing regions of the world that stand to benefit most from hydroelectricity are often those with the highest sediment yields (see Figure 2 on page 8).[3] In these regions, sustainable hydropower development must involve consideration of sediment management techniques during design, construction and operation.

Sediment impacts on generation

About 0.5% to 1% of the total volume of 6,800 km^3 of water stored in reservoirs around the world is lost annually as a result of sedimentation.[2] As a result, global per capita reservoir storage has rapidly decreased since its peak at about 1980. Current storage is equivalent to levels that existed nearly 60 years ago.2



Loss of reservoir storage reduces flexibility in generation and affects the reliability of water supply. Without storage, hydropower facilities are entirely dependent on seasonal flows. These flows might not occur when energy is needed, eliminating one of the key benefits that hydropower provides over other renewables.

Sediments discharged from an upstream dam in a cascade system can increase tailwater levels, reducing power generation.[1] This would impact the generation potential of all plants in the cascade and increase the possibility of powerhouse flooding.

Sediment impacts on stability

Sediment loads are commonly idealized as a static at-rest soil pressure. The U.S. Bureau of Reclamation's design manual for small dams suggests that sediments be considered equivalent to a fluid with an implied pressure coefficient of about 0.39 and an internal friction coefficient of about 37 degrees.

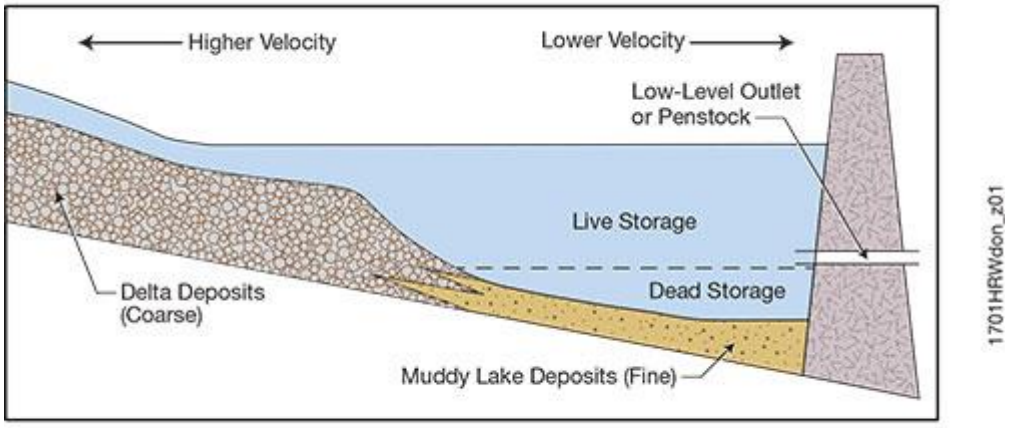
However, actual reservoir sediment properties can vary considerably. Unconsolidated fine-grained sediments likely have lower shear resistance and a higher at-rest pressure coefficient, while a reservoir filled with coarser sediments may have a higher shear strength.1

Published criteria with respect to potential changes in uplift pressures due to sedimentation often neglect the fact that fine-grained sediments may reduce uplift in the same manner as does an engineered upstream blanket. Conversely, in the case where there is a large turbid inflow, higher uplift pressures would be expected until enough particles had settled to form a blanket.

During a seismic event, it is likely that liquefied sediments would quickly return close to their original state, resulting in a rapid dissipation of pore pressures. Therefore, it may be questionable to automatically assign higher uplift pressures in this case.

Commonly used design considerations can omit some plausible load cases. For example, an underwater sediment slope failure could cause surface waves, adding additional loading, hydro-dynamic pressure waves and an inertial loading from the dense fluidized soil-water mass. Another phenomena commonly ignored relates to turbidity currents in reservoirs. Such turbid fluid with a sediment load of 100 mg/l could be about 6% heavier than clear water.1

Figure 1 — Typical Reservoir Sediment Profile*



Typically, sedimentation in the reservoir behind a dam takes the form of progressively finer materials being deposited as the flows approach the dam.

*Adapted from Morris, G.L. and J. Fan, *Reservoir Sedimentation Manual*, McGraw-Hill, New York, 1998.

Submarine landslides are widely studied because of their potential to create tsunami waves. However, designers also need to consider the potential that failure of the steeply sloped deltaic front could increase loading and produce compression waves that may fluidize finer

sediments near the toe of the landslide. As the deposition advances toward the dam, the potential for issues progressively increases.

It is often assumed that, during an earthquake, sediments fully liquefy, lose all strength and exert a dense fluid hydrostatic load on the dam. However, this degree of fluidization likely is not possible in a reservoir filled with coarse materials. Designers also often assume that the fully fluidized dense fluid contributes to hydro-dynamic pressure loading based on Westergaard's formula, ignoring the physical basis for its derivation. In fact, there is some question about the applicability of Westergaard's formula for hydro-dynamic pressures.

Designs also need to consider the degree of saturation of the sediments. There is minimal system damping under dynamic loading when reservoir sediments are fully saturated. However, significant reductions in acceleration occur when sediments are partially saturated.4,5 For rigid foundations, hydrodynamic pressures decrease slightly at the dam base when sediments are fully saturated but increase when partially saturated.[4] Partial saturation will increase he system's response to horizontal ground movement.[5] Sediment thickness is an important consideration, especially when the sediments are partially saturated.[5] Thin layers result in minimal absorption of horizontal motions, largely due to a relatively high modulus of elasticity and low attenuation coefficient.[6] Over the reservoir life cycle, this changes as sediments continue to accumulate.[7] Other important factors are sediment density, compressibility and pore water pressure.[5,7]

This dependence on sediment properties makes a strong case for their measurement and inclusion as part of the design.[4] However, designs are performed before sedimentation occurs and the same sediments that are stable under normal conditions and absorb energy at the bottom of the reservoir could liquefy. For this reason, the use of a reservoir bottom reflection coefficient must be logically linked to assessment of the reservoir sediment behavior and ongoing monitoring.

Sediment impacts on discharge capability

Sediments will often block low-level outlets designed to allow for reservoir drawdown.[1] As sedimentation continues, clogging of spillway tunnels or other conduits may occur.[1]

Reduction of spillway capacity can occur as a result of the loss of approach depth when the sediment front reaches the dam. The reservoir becomes a delta-filled valley that takes a meandering course such that a flood wave does not spread out to allow flood routing.

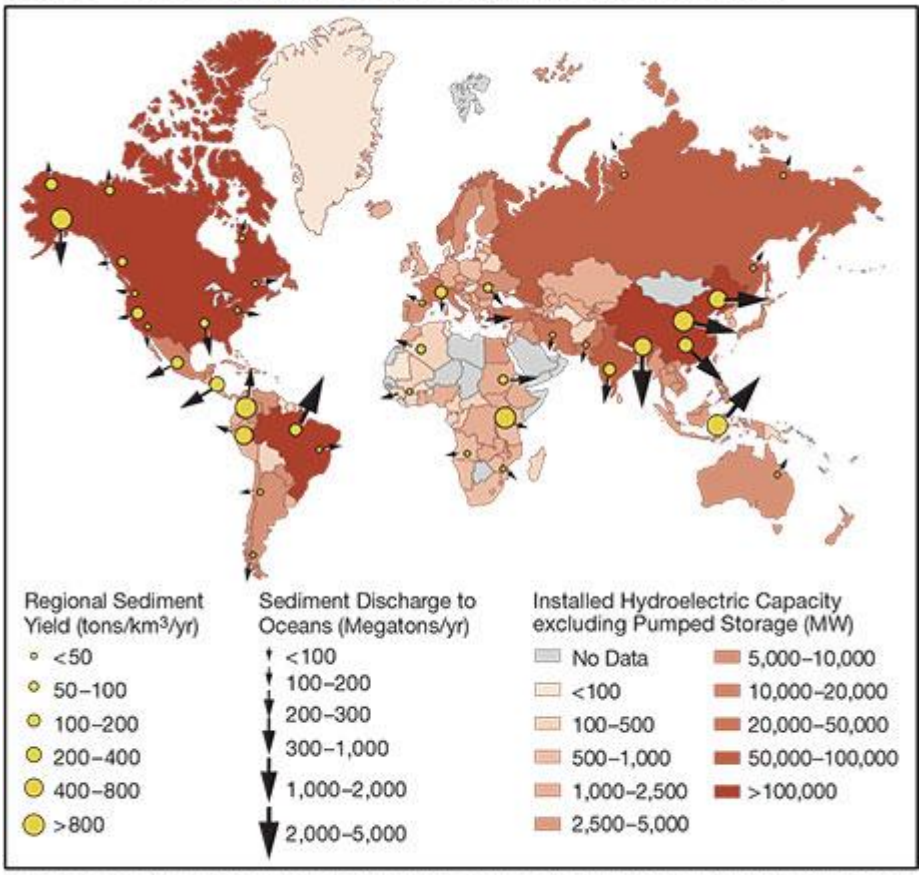
Sediment impacts on equipment

Sediment can damage turbines and other mechanical equipment through erosion of the oxide coating on the blades, leading to surface irregularities and more serious material damage.[8] Sustained erosion can lead to extended shutdown time for maintenance or replacement.[8]

Many factors determine rates of mechanical abrasion. Of particular importance is sediment type and physical characteristics. Angular sediments composed of minerals with a Mohs hardness greater than 5 – such as quartz, feldspar and tourmaline – are problematic. In addition, hydraulic and facility operation parameters such as flow velocity, hydraulic head, turbulence, turbine rotation speed and turbine material affect abrasion susceptibility. Impulse

turbines, such as Pelton or Turgo, are more susceptible to abrasion than are reaction turbines.[8] However, runner changes and needle tip/seat ring replacement are much easier with Pelton turbines. Therefore, they may be preferable on the basis of the overall life cycle cost.

Figure 2 — Comparison of Hydroelectric Potential and Sediment Production*



Developing regions of the world that stand to benefit most from production of hydroelectricity are often those with the highest sediment yields.

*Installed capacity data and sediment yield data adapted from International Hydropower Association 2015 Hydropower Status Report, figure adapted[16]

Abrasion can be reduced by selecting metals to increase erosive resistance and/or by reducing the volume of fine sediment that reaches mechanical equipment. Plants often are designed to remove most of the coarse sediment particles. However, even silt can cause significant abrasion if the quartz content and pressure head is high enough.[9] The 1,500 MW Nathpa Jhakri hydroelectric plant in India used four desilting chambers that were successful in removing coarser sediments. However, damage from the finer particles was so severe that parts of the turbines had to be replaced within one year.

Materials used commonly in sediment-prone hydropower plants are stainless steels that are heat treated for hardening and increased protection from abrasion.[8] Protecting mechanical equipment from sediment abrasion can also be achieved with hard surface coatings of ceramic paints or pastes or with hard facing alloys.[8] Research has shown improved resistance to sediment abrasion when tungsten carbide-based composites are used as a surface coating.[8] In undertaking such assessments, it is important to consider the fact that abrasion will increase as the reservoir fills. The Nozaki method can be used to assess turbine repair frequency. The method accounts for the effective sediment concentration, particle size and shape, the turbine material and any coatings.

Turbine designs need to minimize peak velocities to reduce impacts. For a Pelton turbine, fewer jets and larger runner buckets with larger radii reduce centrifugal forces between the sediment and runner surfaces. Regardless of the turbine selected, designs must consider issues such as the ease of runner removal for future maintenance.

Sediment impacts on the environment

Any dam will cause some degree of sediment starvation downstream. Plant and animal species are sensitive to alteration of both the sediment supply and flow regime.2,10 Increases in sediment concentration can create turbid waters with a smaller euphotic zone. This decreases plant productivity, negatively impacting fish and bird species2 and causing abrasion of fish gills, thus increasing potential for disease or mortality. Turbidity can also cause visual impairment for predatory fish, affecting their feeding habits. Finally, sediment is a primary carrier of suspended pollutants such as nitrogen, phosphorous and heavy metals.10

Sediments released as a result of sediment management or a dam breach may have environmental effects that can persist for decades.

Numerical modeling of sedimentation and sediment management strategies

A variety of tools are available for hydromorphological simulation in order to optimize reservoir management:

- U.S. Army Corps of Engineers' HEC-RAS model features a movable boundary sediment transport calculation module that was recently used to simulate sedimentation processes resulting from hydropower development in northern Manitoba.11
- MIKE 21 is a two-dimensional hydrodynamic model used to simulate sedimentation processes that was used to assess sediment deposition patterns and simulate the results of future flushing operations at Boegoeberg Dam in South Africa.12
- The hydrodynamic, sediment transport and physical habitat model FAST is used to simulate morphological processes and changes to fish habitat within alluvial rivers.13 It was used to predict hydromorphological conditions and to optimize sediment flushing procedures prior to constructing new hydropower facilities on the Nile River.

Sediment management solutions

Developing and retaining sustainable storage to satisfy global needs requires inclusion of reservoir sediment management practices at project conception and throughout its life cycle. These practices vary depending on the tope of facility. For run-of-river projects, sediment management aims to remove sediments that can cause abrasion of the turbines and clog cooling water intakes. In a storage project, this objective and extending reservoir longevity are key.

For storage hydro, sediment management strategies to extend reservoir longevity can be classified into three categories:

- Those that divert some of the sediment through or around the reservoir;
- Those that remove or rearrange sediment that has already been deposited; and

- Those that minimize the amount of sediment reaching the reservoir from upstream.

Many dam operators have implemented sediment management techniques designed to achieve these goals.14 Some examples are described below.

Bypassing

On-stream sediment bypassing diverts part of the sediment-laden water around the reservoir, typically using a weir that operates during high flows when sediment concentrations are high.

An off-stream reservoir can be used such that only the clear water is diverted over a bypass weir. An off-stream reservoir typically has limited capacity and can only exclude sediments carried by higher streamflows.[2] However, it does reduce the amount of suspended sediment and bedload reaching the reservoir.14 Other advantages include the fact that the reservoir and dam are located away from the main river channel, allowing for minimal disruption to aquatic species and habitat and reducing the need for large on-stream spillways.[2] On the other hand, off-stream reservoirs typically do not permit maximization of generation capacity, especially in areas that depend on high stream flows occurring over a short period of time.[2]

Sediment bypassing works best in areas of high relief where the sediment-laden flows are carried efficiently through the diversion tunnel or channel. Bypassing is most cost-effective at dams that are on the bend of a river, as this allows for a relatively short diversion between the weir and the downstream side of the dam.14

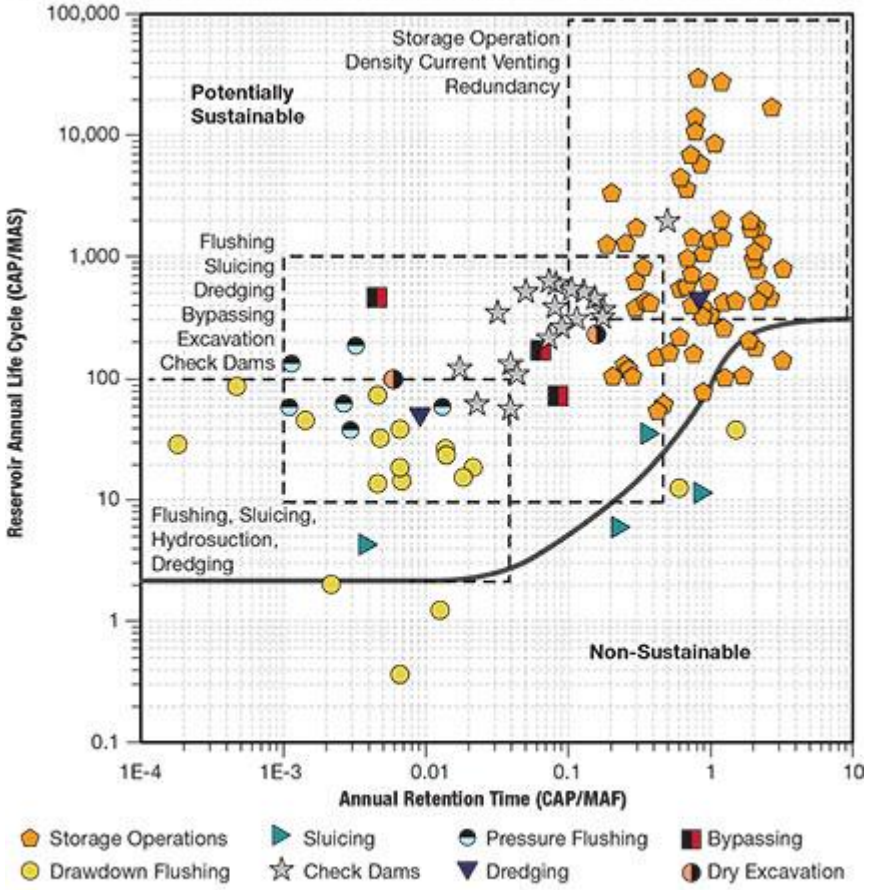
Sluicing/drawdown routing

This technique involves lowering the reservoir water levels in advance of high streamflows so that water and sediment can be routed through the spillway at high velocities. Refill occurs during the receding limb of the flood hydrograph.2,14 Sluicing methods depend on the facility's hydrologic characteristics and reservoir size.

Dredging

Dredging can be efficient but it will continue for the life of the project and can have significant cost impacts. For example, dredging of 6 million m^3 of sediment at the Loiza reservoir in Puerto Rico in 1997 cost \$10/m^3.1,2

Figure 3 — Sediment Management Techniques Related to Reservoir Life and Retention Time*



The selection of the optimal sediment management technique for a reservoir can be estimated based on precedent experience and other factors, such as the ratio of reservoir volume to mean annual sediment inflow and retention time.

*Adapted[9]

Flushing

Flushing involves emptying the reservoir by opening bottom outlets and allowing the incoming streamflow to scour sediment.2,14 The effectiveness varies but, generally, only a "core" of sediment along the original channel thalweg is flushed. Sediments on the sides of the reservoir remain in place.2

An alternative method is pressure flushing, where the reservoir is partially drawn down before flushing. This redistributes coarse upstream sediments closer to the dam, alleviating their impacts, but often does not clear the finer sediments.[2] Pressure flushing is also used for sediment redistribution, moving them to a less sensitive location.

Erosion control

Many watersheds experience increased erosion rates due to land use and other human practices. Erosion reduction techniques fall into three categories: structural or mechanical, vegetative and operational.1

Structural or mechanical measures – such as terraces, conveyance channels, check dams and sediment traps1,14 – decrease overland or channelized flow velocity, increasing surface storage and thereby reducing the sediment load in the runoff.

Vegetative erosion control takes advantage of plants' natural ability to limit erosion. Agricultural practices that minimize sediment yield are particularly effective.

Operational measures minimize erosion through planning, management and organization. Examples include timing construction work such that erosion is minimized or scheduling timber harvesting to coincide with favorable soil conditions.1

Erosion management is perhaps the most widely recommended but most poorly implemented sediment management technique because land users may not see any direct benefits from controlling sediment yield.

Sedimentation can cause a number of problems, including:

- Reduced water storage: Sediment can fill up space that would otherwise be used for water storage.
- Reduced recreational access: Sediment can make reservoirs shallower, reducing access for recreational activities.
- Impacted hydropower: Sediment can collect around hydropower installations, making it difficult for them to operate.
- Increased dam pressure: Sediment can add pressure to the dam.
- Harmed aquatic life: Sedimentation can harm aquatic life.
- Coastal erosion: Dredging can alter the natural shape and sedimentation of a waterbody, which can contribute to coastal erosion.

Some ways to manage sedimentation include:

- Flushing

  Flushing reservoirs and regulated rivers more frequently can help deliver sediment downstream, where it can be beneficial for the river's health.

- Sediment bypass systems

  During major flood events, sediment bypass systems can reroute sediment-laden waters to prevent them from entering the reservoir.

- Sluicing
  Sluicing, or sediment routing operations, can help prevent sediment from depositing in reservoirs.

India faces a number of sedimentation problems, including:

- Reservoir sedimentation

  Sedimentation in Indian reservoirs is a serious problem due to a number of factors, including:

  - Population growth

    - Deforestation
    - Poor soil conservation
    - Intense agricultural operations
    - High rainfall events
    - Geological and anthropogenic activities
- Flooding

  Deforestation in headwater areas can cause frequent flooding in the Gangetic plains.
- Hydropower projects

  Sedimentation can cause problems for hydropower projects, including:

    - Abrading turbine blades
    - Damaging dams and other structures
    - Reducing reservoir storage
- Water security

  If sedimentation is not prevented, India's water security could be at risk.

Sedimentation can be prevented, managed, and reduced. This can help to ensure that more water is stored, which is important for a sustainable future.

Sedimentation can cause a number of environmental problems, including:

- Water pollution: Sediment can make water cloudy, which can prevent animals from seeing food and prevent natural vegetation from growing.
- Habitat degradation: Sediment can bury and suffocate fish eggs, smother invertebrates, and fill in spaces where animals live. This can reduce the amount of habitat and spawning grounds for fish.
- Food chain disruption: Sediment can destroy the habitat of the smallest organisms in a stream, which can lead to a decline in fish populations.
- Marine ecosystem damage: Sediment can alter sea floor habitats, clog the gills of filter feeders, and poison marine life.
- Reduced water clarity: Sediment can reduce the clarity of water in coastal areas.
- Reduced water flow: Sediment can fill and embed gravel and cobble habitats, which can reduce water and oxygen flow.
- Toxic compounds: Sediment particles can carry toxic agricultural and industrial compounds.

Sediment forms when bedrock is turned into smaller particles through weathering. Weathering can be mechanical, such as pressure expansion, frost wedging, and root wedging, or chemical, such as carbonic acid and hydrolysis.

Lecture No 5

here are several corrective measures for erosion, including:

- Vegetation

  Planting trees, grasses, legumes, and other vegetation can help stabilize soil and prevent erosion. Trees can help prevent landslides on steep slopes and stream bank erosion, but they may not be effective on moderately sloping hillslopes.

- Mulch

  Mulch can help protect soil from erosion, especially on footpaths with exposed soil.

- Check dams and terraces

  Check dams and terraces are effective for heavy erosion in areas with concentrated flow.

- Erosion control blankets

  These blankets can be made from natural or synthetic materials and slow down the flow of water.

- Silt fencing

  Silt fencing traps sediment as water flows through it.

- Energy dissipaters

  These structures absorb the energy of flowing water.

- Articulated concrete blocks

  These blocks can be used to control erosion on artificial drainage channels and embankments of waterways.

- Mathematical modeling

  Mathematical models can help understand the mechanisms of soil erosion and resulting sediment surface runoff.

- Stormwater redirection

  Redirecting stormwater and roof runoff to areas that can settle and dissipate water can help.

- Crop rotation
  Crop rotation can help reduce soil erosion by fixing the land with alternatively planted deep-rooted crops.

Here are some corrective measures for sedimentation:

- Sediment control

Temporary barriers like silt fences, berms, and sedimentation ponds can reduce runoff velocity and allow sediment to settle.

- Erosion control

  Maintaining healthy vegetation can reduce the amount of sedimentation transported by runoff.

- Soil stabilization

  Reducing erosion in borrow and waste areas, cut and fill slopes, and road shoulders can reduce sedimentation.

- Sediment traps

  A sediment trap is a section of the estuarine bed that is deepened to a greater depth than its surroundings.

- Hydraulic flushing

  A popular and cost-effective method for sedimentation control in sewers. It involves releasing a large volume of water quickly to create a flush wave that flushes out sediment.

- Sediment removal

  Flushing and sluicing are techniques that can remove or minimize sediment deposition.

- Adaptive strategies
  These actions mitigate the impacts of sedimentation without handling the sediment. For example, raising a dam or applying a protective coating to increase equipment resistance.

Lecture No 6

Biological disaster

A natural catastrophe having unfortunate consequences is called a disaster. Biological disasters are natural scenarios involving disease, disability, or death on a large scale among humans, animals, and plants due to micro-organisms like bacteria, or viruses, or toxins.

In this article, you can read all about biological disaster management in India for IAS Exam. Disaster management is an important topic in the UPSC syllabus and one that is featured regularly in the newspapers. Here, you can read about the term, types, and the institutional and legal frameworks in India.

What is a Biological Disaster?

Definition	The devastating effects caused by an enormous spread of a certain kind of living organism that may spread disease, viruses or infestation of plant, animal or insect life on an epidemic or pandemic level
Epidemic Level	Indicates a disaster that affects many people in a given area or community
Pandemic Level	Indicates a disaster that affects a much larger region, sometimes an entire continent or even the whole planet. For example, the recent H1N1 or Swine Flu pandemic

Biological Disasters

1. The nodal agency for handling epidemics – Ministry of Health and Family Welfare
    - Decision-making
    - Advisory body
    - Emergency medical relief providing
2. The primary responsibility of dealing with biological disasters is with the State Governments. (Reason – Health is a State Subject).
3. The nodal agency for investigating outbreaks – **National Institute of Communicable Diseases (NICD)**
4. Nodal ministry for Biological Warfare – Ministry of Home Affairs (Biological warfare is the use of biological agents as an act of war)

Biological Disasters – Classifications

Charles Baldwin developed the symbol for biohazard in 1966.



The US Center for Disease Control classifies biohazards into four biosafety levels as follows:

1. **BSL-1**: Bacteria and Viruses including Bacillus subtilis, some cell cultures, canine hepatitis, and non-infectious bacteria. Protection is only facial protection and gloves.
2. **BSL-2**: Bacteria and viruses that cause only mild disease to humans, or are difficult to contract via aerosol in a lab setting such as hepatitis A, B, C, mumps, measles, HIV, etc. Protection – use of autoclaves for sterilizing and biological safety cabinets.
3. **BSL-3**: Bacteria and viruses causing severe to fatal disease in humans. Example: West Nile virus, anthrax, MERS coronavirus. Protection – Stringent safety protocols such as the use of respirators to prevent airborne infection.
4. **BSL-4**: Potentially fatal (to human beings) viruses like Ebola virus, Marburg virus, Lassa fever virus, etc. Protection – use of a positive pressure personnel suit, with a segregated air supply.

Biological Disaster – Legislations

The following legislations have been enacted in India for the prevention of biohazards and implementation of protective, eradicative, and containing measures when there is an outbreak:

1. The Water (Prevention and Control of Pollution) Act, 1974
2. The Air (Prevention and Control of Pollution) Act, 1981
3. The Environmental (Protection) Act, 1986 and the Rules (1986)
4. Disaster Management Act 2005 – this provides for the institutional and operational framework for disaster prevention, mitigation, response, preparedness, and recovery at all levels.

Prevention of Biological Hazards

The basic measure to prevent and control biohazards is the elimination of the source of contamination.

Preventive Measures for workers in the field (medical):

1. Engineering controls – to help prevent the spread of such disasters including proper ventilation, installing negative pressure, and usage of UV lamps.
2. Personal hygiene – washing hands with liquid soap, proper care for clothes that have been exposed to a probably contaminated environment.
3. Personal protection equipment – masks, protective clothing, gloves, face shield, eye shield, shoe covers.
4. Sterilization – Using ultra heat or high pressure to eliminate bacteria or using biocide to kill microbes.
5. Respiratory protection – surgical masks, respirators, powered air-purifying respirators (PAPR), air-supplying respirators.

Prevention of Biological Disasters

Prevention includes the following measures that should be taken before (preventive), during, and after any outbreaks.

Environmental Management:

1. Safe water supply, proper maintenance of sewage pipelines – to prevent waterborne diseases such as cholera, typhoid, hepatitis, dysentery, etc.
2. Awareness of personal hygiene and provision for washing, cleaning, bathing, avoiding overcrowding, etc.
3. Vector control:
    - Environmental engineering work and generic integrated vector control measures.
    - Water management, not permitting water to stagnate and collect and other methods to eliminate breeding places for vectors.
    - Regular spraying of insecticides, outdoor fogging, etc. for controlling vectors.
    - Controlling the population of rodents.

Post-disaster Epidemics Prevention:

1. The risk of epidemics is increased after any biological disaster.
2. Integrated Disease Surveillance Systems (IDSS) monitors the sources, modes of diseases spreading, and investigates the epidemics.

Detection and Containment of Outbreaks:

This consists of four steps as given under:

1. Recognizing and diagnosing by primary healthcare practitioners.
2. Communicating surveillance information to public health authorities.
3. Epidemiological analysis of surveillance data
4. Public health measures and delivering proper medical treatment.

Biological Disaster – Legal Framework

1. The Epidemic Diseases Act was enacted in the year 1897.
2. This Act does not provide any power to the center to intervene in biological emergencies.
3. It has to be substituted by an Act that takes care of the prevailing and foreseeable public health needs including emergencies such as BT attacks and the use of biological weapons by an adversary, cross-border issues, and international spread of diseases.
4. It should give enough powers to the central and state governments and local authorities to act with impunity, notify affected areas, restrict movement or quarantine the affected area, enter any premises to take samples of suspected materials, and seal them.
5. The Act should also establish controls over biological sample transfer, biosecurity, and biosafety of materials/laboratories.

Biological Disaster – Institutional Framework

In the Ministry of Health & Family Welfare (MoH&FW), public health needs to be accorded high priority with a separate Additional Directorate General of Health and Sanitation (DGHS) for public health. In some states, there is a separate department of public health. States that do not have such arrangements will also have to take initiatives to establish such a department.

Biological Disaster – Operational Framework

1. At the national level, there is no policy on biological disasters. The existing contingency plan of MoH&FW is about 10 years old and needs extensive revision. All components related to public health, namely apex institutions, field epidemiology, surveillance, teaching, training, research, etc., need to be strengthened.
2. At the operational level, Command and Control (C&C) are identifiable clearly at the district level, where the district collector is vested with certain powers to requisition resources, notify a disease, inspect any premises, seek help from the Army, state, or center, enforce quarantine, etc. However, there is no concept of an incident command system wherein the entire action is brought under the ambit of an incident commander with support from the disciplines of logistics, finance, and technical teams, etc. There is an urgent need for establishing an incident command system in every district.
3. There is a shortage of medical and paramedical staff at the district and sub-district levels. There is also an acute shortage of public health specialists, epidemiologists, clinical microbiologists, and virologists.
4. Biosafety laboratories are required for the prompt diagnosis of the agents for the effective management of biological disasters. There is no BSL-4 laboratory in the human health sector. BSL- 3 laboratories are also limited. Major issues remain regarding biosecurity, the indigenous capability of preparing diagnostic reagents, and quality assurance.

5. Lack of an Integrated Ambulance Network (IAN). There is no ambulance system with advanced life-support facilities that are capable of working in biological disasters.
6. State-run hospitals have limited medical supplies. Even in normal situations, a patient has to buy medicines. There is a lack of stockpile of drugs, important vaccines like the anthrax vaccine, PPE, or diagnostics for surge capacity. In a crisis, there is further incapacitation due to tedious procurement procedures.

Biological disasters are scenarios involving disease, disability or death on a large scale among humans, animals and plants due to toxins or disease caused by live organisms or their products. Such disasters may be natural in the form of epidemics or pandemics of existing, emerging or reemerging diseases and pestilences or man-made by the intentional use of disease causing agents in Biological Warfare (BW) operations or incidents of Bioterrorism (BT).

It might be caused by epidemics, accidental release of virulent microorganism(s) or Bioterrorism (BT) with the use of biological agents such as anthrax, smallpox, etc. The existence of infectious diseases have been known among human communities and civilizations since the dawn of history. The classical literature of nearly all civilizations record the ability of major infections to decimate populations, thwart military campaigns and unsettle nations. Social upheavals caused by epidemics have contributed in shaping history over the ages. The mutual association of war, pestilence and famine was acknowledged and often attributed to divine influences, though a few keen observers realized that some infections were contagious. The development of bacteriology and epidemiology later, established the chain of infection. Along with nuclear and chemical agents, which are derived from technology, biological agents have been accepted as agents of mass destruction capable of generating comparable disasters.

Biological hazards and disasters are events that can cause harm to the health of humans, animals, and plants due to harmful biological agents. These agents can include bacteria, viruses, toxins, or other organic substances. Biological disasters can be natural or man-made.

Some examples of biological hazards include:

- Epidemics: Diseases that affect a large number of people in a community or region at the same time. Cholera and plague are examples of epidemics.
- Pandemics: Epidemics that spread across a large region, such as a continent or worldwide. Influenza H1N1 (Swine Flu) is an example of a pandemic.
- Venomous wildlife and insects: These can pose a threat to human health.
- Poisonous plants: These can pose a threat to human health.
- Mosquitoes carrying disease-causing agents: These can pose a threat to human health.

If you are exposed to a biological hazard, you can take these precautions: Move away quickly, Wash with soap and water, Contact authorities, Listen to the media for official instructions, and Seek medical attention if you become sick.

Lecture No 7

Population Explosion

According to the Census Act 1948, for every ten years, the census is conducted in India to count the total population in our country.

The last census was conducted in the previous year in 2021, and the next census will take place in 2031.

In terms of population, India ranks in second position.

The government takes various measures in India to reduce the speed of population explosion.

In this chapter, let us learn more about population, population explosion, effects of increasing population and various measures taken to control population growth.

What is Population?

A group of all living organisms of a particular species existing in a specific geographical area, at one particular time with the capability of interbreeding, is called population.

According to the 2021 census, the current population of India is 1.4 billion.

What is Population Explosion?

A rapid increase in the population is known as a population explosion.

The two causes of population explosion are:

- Increase in the birth rate.
- Decrease in the death rate.

An increase in population is directly related to the birth rate. The population in India is rising at a considerable speed, which results in various problems, including social, economic and environmental.

Effects of Increasing Population

Increasing population has many adverse effects on our environment.

Listed below are the significant effects of increasing population:

- Increase in slum areas.
- Rise in unemployment and underemployment.

- Eradicating poverty and hunger becomes more difficult.
- Providing better health and education facilities becomes more difficult.
- This can lead to less availability of land, water, food and other necessary resources.
- An increase in population may lead to the unavailability of food grains due to the excessive demands of people.
- Overexploitation of water and other natural resources results in no scope of replenishment, which is a recipe for a natural disaster.
- Increasing population increases air and water pollution, leading to an increase in diseases, which ultimately leads to a rise in expenditure on healthcare.

Measures to Control the Population Explosion

Control measures can be broadly classified into the direct method and indirect method.

The direct method includes:

- Focus on family planning measures.
- Providing free education and advice to the people.
- Strictly follow the minimum age criteria of marriage.

The indirect method includes:

- Literacy.
- Female empowerment.
- Improvement in quality of life.
- Positive attitudes towards women’s rights in sexual relationships.

Lecture No 8

Man-made disasters
Human-induced disasters are very dangerous catastrophes caused by humans. They are caused by human activity. The following are the examples:
Chemical spills, hazardous material spills, explosives, chemical or biological attacks, nuclear blasts, rail accidents, airline crashes, or groundwater poisoning are all instances of man-made disasters.

Man-made hazards and disasters are caused by human error, negligence, or intent, and can include:

- Crime: Arson, civil disorder, and terrorism
- Biological and chemical threats: Chemical spills, hazardous material spills, and biological attacks
- Cyber-attacks: Disruptions to communications
- Transportation accidents: Airline crashes, rail accidents, and road accidents
- Industrial accidents: Power outages
- Nuclear explosions and radiation: Nuclear material used for peaceful or military purposes
- Hazardous materials: Anything that can be harmful if not used properly, such as solids, liquids, or gases
- Catastrophic climate change: A combination of natural and human factors
- Bioterrorism: A man-made disaster

Other examples of man-made hazards include: Boating, Fire, Crowd crushes, and Weapons of mass destruction.

Man-made hazards are events that are caused by human error, negligence, or intent, and can be life-threatening or inconvenient:

- Hazardous materials: These can be released through accidents in transportation or chemical plants.
- Power outages: These can be caused by accidents or weather.
- Terrorism: This is an intentional violent act that can be used to achieve political or ideological goals.
- Boating accidents: A large pleasure craft hitting rocks can release a large amount of fuel into the environment.
- Fires: Fires can spread quickly and become life-threatening in a short amount of time.

- Nuclear accidents: Nuclear material can be disastrous if not handled properly.
- Biological agents: These can be a man-made hazard.
- Disruptions in services: These can include disruptions to water, sewer, communications, and travel.
- Weapons of mass destruction: These can be a man-made hazard.

In the event of a major disaster, there can be many fatalities and injuries, large groups of people displaced, and difficulty providing food, clothing, and shelter